getops:

```
srivatsan@MSI:~/linux-practice$ cat > employee_info.sh <<'EOF'
#!/bin/bash
while getopts "n:d:" opt; do
    case "$opt" in
        n) name="$OPTARG" ;;
        d) department="$OPTARG" ;;
        *) echo "Invalid option"; exit 1 ;;
    esac
done

echo "Employee: $name"
echo "Department: $department"
EOF
srivatsan@MSI:~/linux-practice$ chmod +x employee_info.sh
srivatsan@MSI:~/linux-practice$ ./employee_info.sh -n Alice -d HR
Employee: Alice
Department: HR
srivatsan@MSI:~/linux-practice$ ./employee_info.sh -n Bob -d IT
Employee: Bob
Department: IT
srivatsan@MSI:~/linux-practice$
```

mapfile:

```
srivatsan@MSI:~/linux-practice$ mapfile -t employees < names.txt
srivatsan@MSI:~/linux-practice$ echo "${employees[@]}"
Alice Bob Charlie David Eva
srivatsan@MSI:~/linux-practice$ echo "${employees[0]}"
Alice
srivatsan@MSI:~/linux-practice$ echo "${employees[2]}"
Charlie
srivatsan@MSI:~/linux-practice$ echo "${#employees[@]}"
5
srivatsan@MSI:~/linux-practice$
```

namei:

```
srivatsan@MSI:~/linux-practice$ namei -l ~/linux-practice/employees.txt
f: /home/srivatsan/linux-practice/employees.txt
drwxr-xr-x root      root      /
drwxr-xr-x root      root      home
drwxr-x--- srivatsan srivatsan srivatsan
drwxr-xr-x srivatsan srivatsan linux-practice
-rw-r--r-- srivatsan srivatsan employees.txt
srivatsan@MSI:~/linux-practice$ namei -l ~/linux-practice
f: /home/srivatsan/linux-practice
drwxr-xr-x root      root      /
drwxr-xr-x root      root      home
drwxr-x--- srivatsan srivatsan srivatsan
drwxr-xr-x srivatsan srivatsan linux-practice
srivatsan@MSI:~/linux-practice$ namei -l /etc/passwd
f: /etc/passwd
drwxr-xr-x root root /
drwxr-xr-x root root etc
-rw-r--r-- root root passwd
srivatsan@MSI:~/linux-practice$
```

pr:

```
srivatsan@MSI:~/linux-practice$ pr -h "EMPLOYEE REPORT" employees.txt

2026-11-05 12:54              EMPLOYEE REPORT                Page 1

Alice HR 25000
Bob IT 35000
Charlie Finance 30000
David HR 28000
Eva IT 40000
```

dos2unix:

```
srivatsan@MSI:~/linux-practice$ file windows.txt
windows.txt: ASCII text, with CRLF line terminators
srivatsan@MSI:~/linux-practice$ dos2unix windows.txt
dos2unix: converting file windows.txt to Unix format...
srivatsan@MSI:~/linux-practice$ file windows.txt
windows.txt: ASCII text
srivatsan@MSI:~/linux-practice$ cat windows.txt
Employee ID
101
102
103
srivatsan@MSI:~/linux-practice$
```

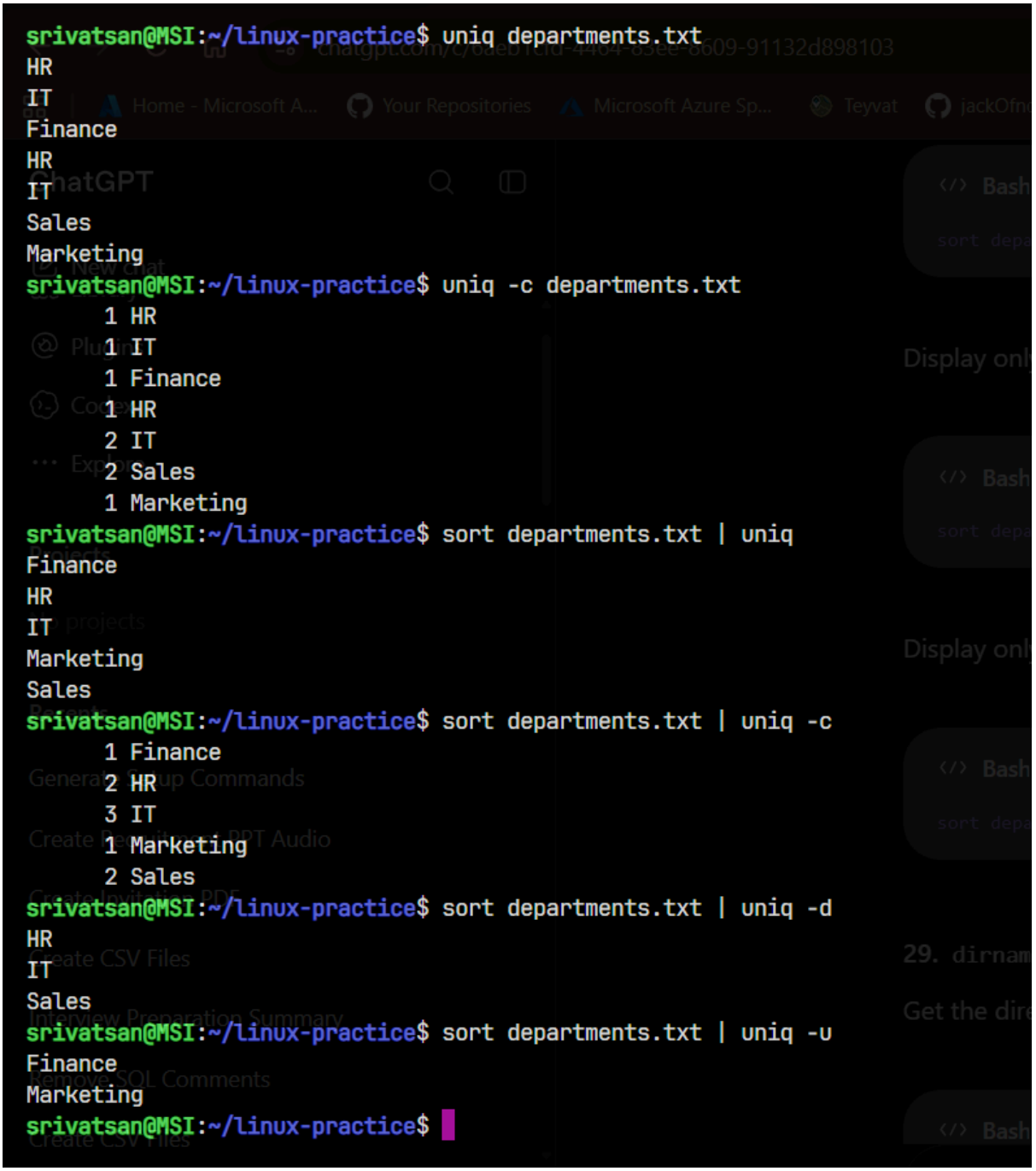
base64:

```
srivatsan@MSI:~/linux-practice$ echo -n "Linux Administration" | base64
TGludXggQWRtaW5pc3RyYXRpb24=
srivatsan@MSI:~/linux-practice$ echo -n "TGludXggQWRtaW5pc3RyYXRpb24=" | base64 -d
Linux Administrationsrivatsan@MSbase64 employees.txt > employees_encoded.txtmployees_encoded.txt
srivatsan@MSI:~/linux-practice$ base64 -d employees_encoded.txt > employees_decoded.txt
srivatsan@MSI:~/linux-practice$ diff employees.txt employees_decoded.txt
srivatsan@MSI:~/linux-practice$ ls
```

tr:

```
srivatsan@MSI:~/linux-practice$ echo "linux administration" | tr 'a-z' 'A-Z'
LINUX ADMINISTRATION
srivatsan@MSI:~/linux-practice$ echo "LINUX COMMANDS" | tr 'A-Z' 'a-z'
linux commands
srivatsan@MSI:~/linux-practice$ echo "Linux Command Practice" | tr ' ' '_'
Linux_Command_Practice
srivatsan@MSI:~/linux-practice$ echo "Employee123Department456" | tr -d '0-9'
EmployeeDepartment
srivatsan@MSI:~/linux-practice$ echo "Linux     Administration" | tr -s ' '
Linux Administration
srivatsan@MSI:~/linux-practice$
```

uniq:


srivatsan@MSI:~/linux-practice$ uniq departments.txt
HR
IT
Finance
HR
IT
Sales
Marketing
srivatsan@MSI:~/linux-practice$ uniq -c departments.txt
1 HR
1 IT
1 Finance
1 HR
2 IT
2 Sales
1 Marketing
srivatsan@MSI:~/linux-practice$ sort departments.txt | uniq
Finance
HR
IT
Marketing
Sales
srivatsan@MSI:~/linux-practice$ sort departments.txt | uniq -c
1 Finance
2 HR
3 IT
1 Marketing
2 Sales
srivatsan@MSI:~/linux-practice$ sort departments.txt | uniq -d
HR
IT
Sales
srivatsan@MSI:~/linux-practice$ sort departments.txt | uniq -u
Finance
Marketing
srivatsan@MSI:~/linux-practice$

dirname:

```
srivatsan@MSI:~/linux-practice$ dirname ~/linux-practice/employees.txt
/home/srivatsan/linux-practice
srivatsan@MSI:~/linux-practice$ FILE_PATH="$HOME/linux-practice/employees.txt"
srivatsan@MSI:~/linux-practice$ DIR_PATH=$(dirname "$FILE_PATH")
srivatsan@MSI:~/linux-practice$ echo "$DIR_PATH"
/home/srivatsan/linux-practice
srivatsan@MSI:~/linux-practice$ dirname /home/user/linux-practice/employees.txt
/home/user/linux-practice
srivatsan@MSI:~/linux-practice$
```

getfacl:

```
srivatsan@MSI:~/linux-practice$ echo "Confidential employee information" > confidential.txt
srivatsan@MSI:~/linux-practice$ getfacl confidential.txt
# file: confidential.txt
# owner: srivatsan
# group: srivatsan
user::rw-
group::r--
other::r--

srivatsan@MSI:~/linux-practice$ sudo setfacl -m u:nobody:r confidential.txt
[sudo: authenticate] Password:
srivatsan@MSI:~/linux-practice$ sudo setfacl -m u:nobody:rw confidential.txt
srivatsan@MSI:~/linux-practice$ getfacl confidential.txt
# file: confidential.txt
# owner: srivatsan
# group: srivatsan
user::rw-
user:nobody:rw-
group::r--
mask::rw-
other::r--

srivatsan@MSI:~/linux-practice$
```